

Embedding Arithmetic: A Lightweight, Tuning-Free Framework for post-hoc Bias Mitigation in Text-to-Image Models

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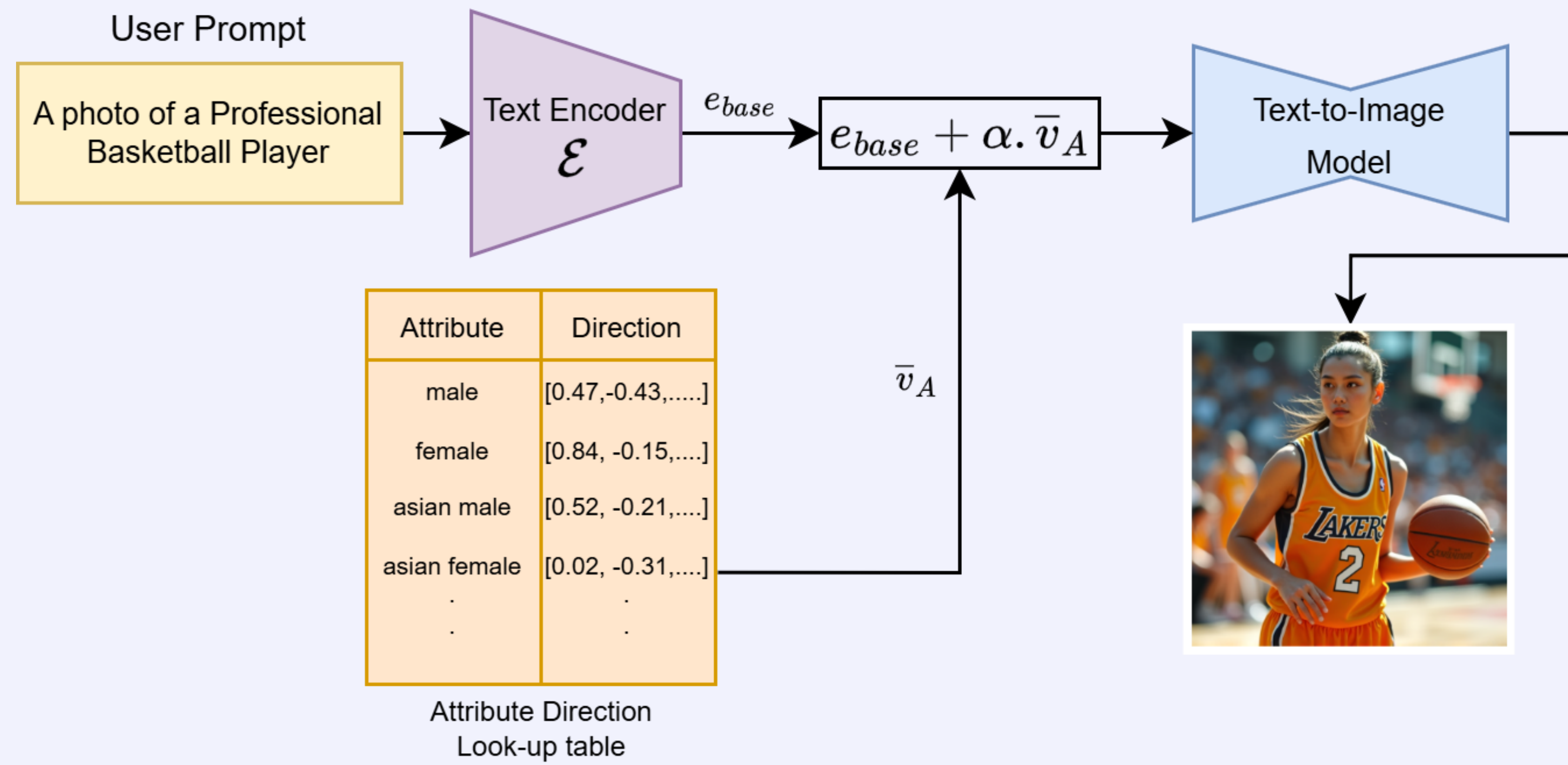
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Introduction

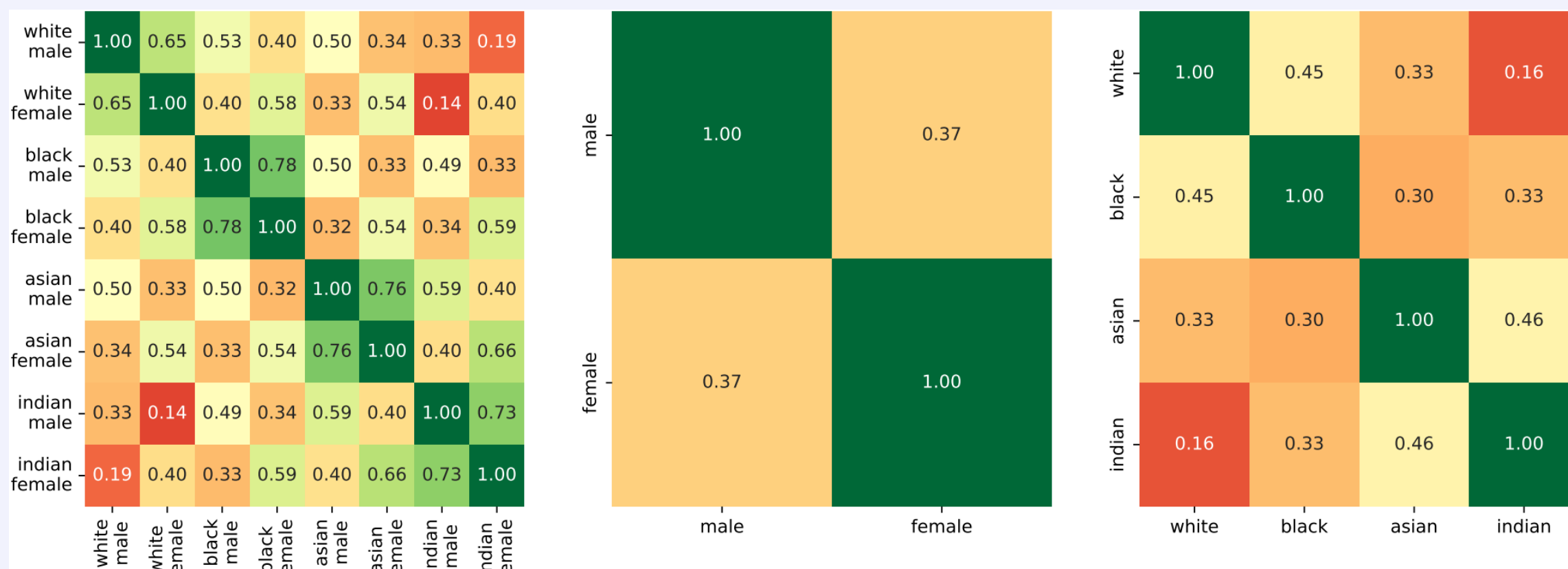
Text-to-Image models amplify harmful societal biases, and existing mitigation methods typically require retraining, fine-tuning, or prompt engineering [1]. We introduce a training-free, inference-time method that corrects bias directly in the conditional embedding space, resolving the fairness-coherence trade-off without altering model weights, prompts, or training data.



Attribute Direction: an attribute vector ($\bar{v}_A$) is computed as the average embedding difference between paired attribute-conditioned prompts (e.g., “a photo of a male doctor” vs “a photo of a doctor”), capturing the direction associated with that attribute in embedding space [2]. **Embedding Arithmetic:** the base prompt embedding (e_{base}) is shifted by $\alpha \cdot \bar{v}_A$ before conditioning the generator, where α controls the strength of the correction. **Look-up table:** pre-computed attribute directions are reused across prompts without recomputation, keeping the method lightweight and tuning-free.

Orthogonality

Hypothesis: If concepts are learned in a disentangled manner, their attribute vectors should be approximately orthogonal.

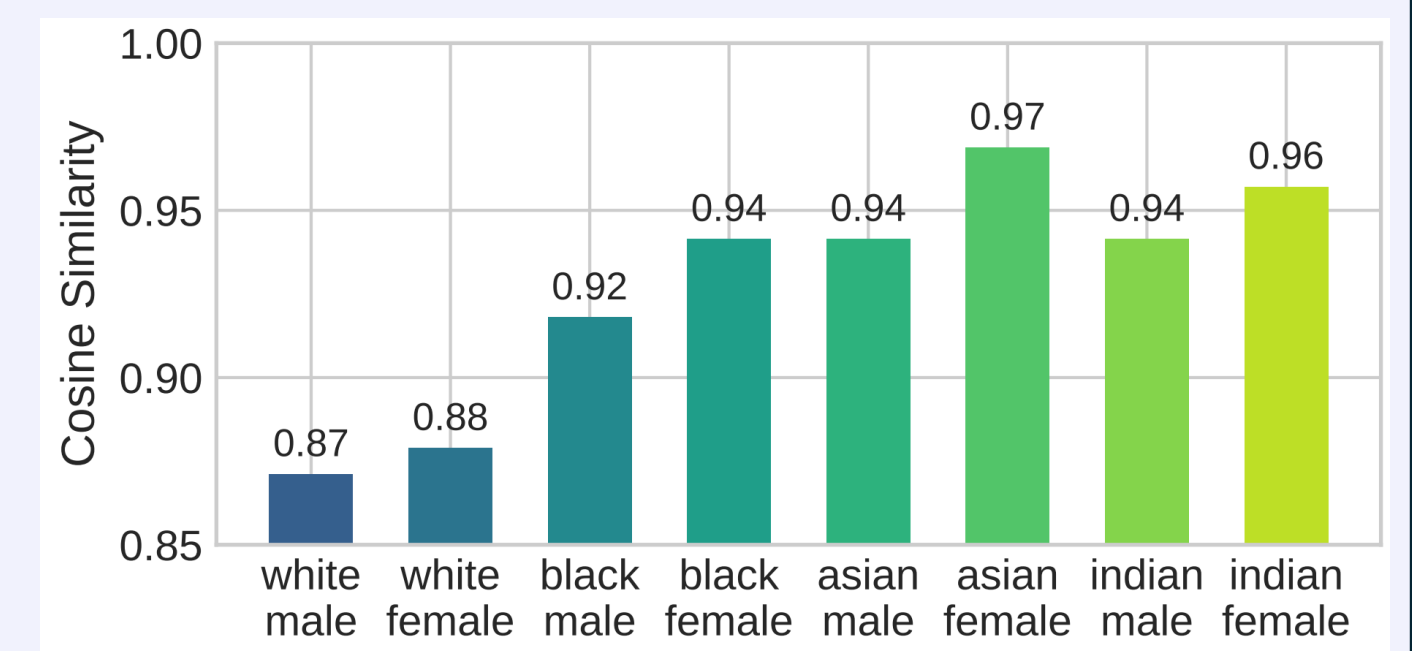


Attribute Directions are not orthogonal: aggregated gender and race vectors show moderate correlation (up to 0.65), and per-identity pairs vary widely in similarity (0.14-0.78) – evidence that the embedding space encodes an entangled, identity-specific structure rather than independent, disentangled axes

Composability

Hypothesis: If the latent space is perfectly compositional, the vector for an intersectional concept should be equivalent to the linear sum of its constituent attribute vectors.

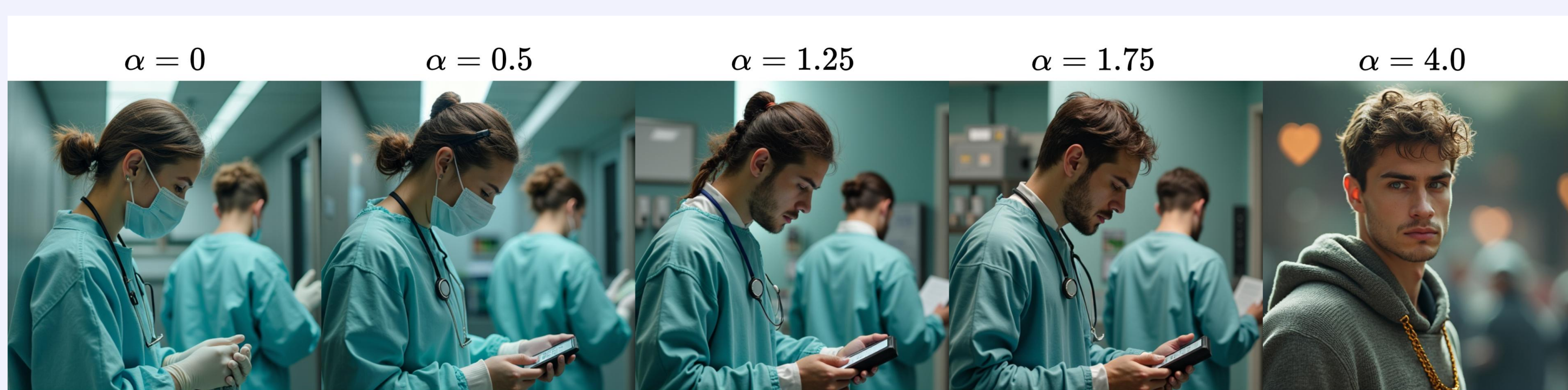
Experiment: To prove this hypothesis, we compare the composed vector $v_{comp} = v_{race} + v_{gender}$ and the intersectional vector $v_{race \oplus gender}$. We then compute their cosine similarity.



Composed attribute vectors remain near-additive (cosine sim 0.87-0.97), confirming approximate compositionality and enabling intersectional steering – e.g., correcting race and gender jointly rather than one attribute at a time. For additional compositionality experiments (more attribute combinations), see the paper’s supplementary material.

Local Linearity

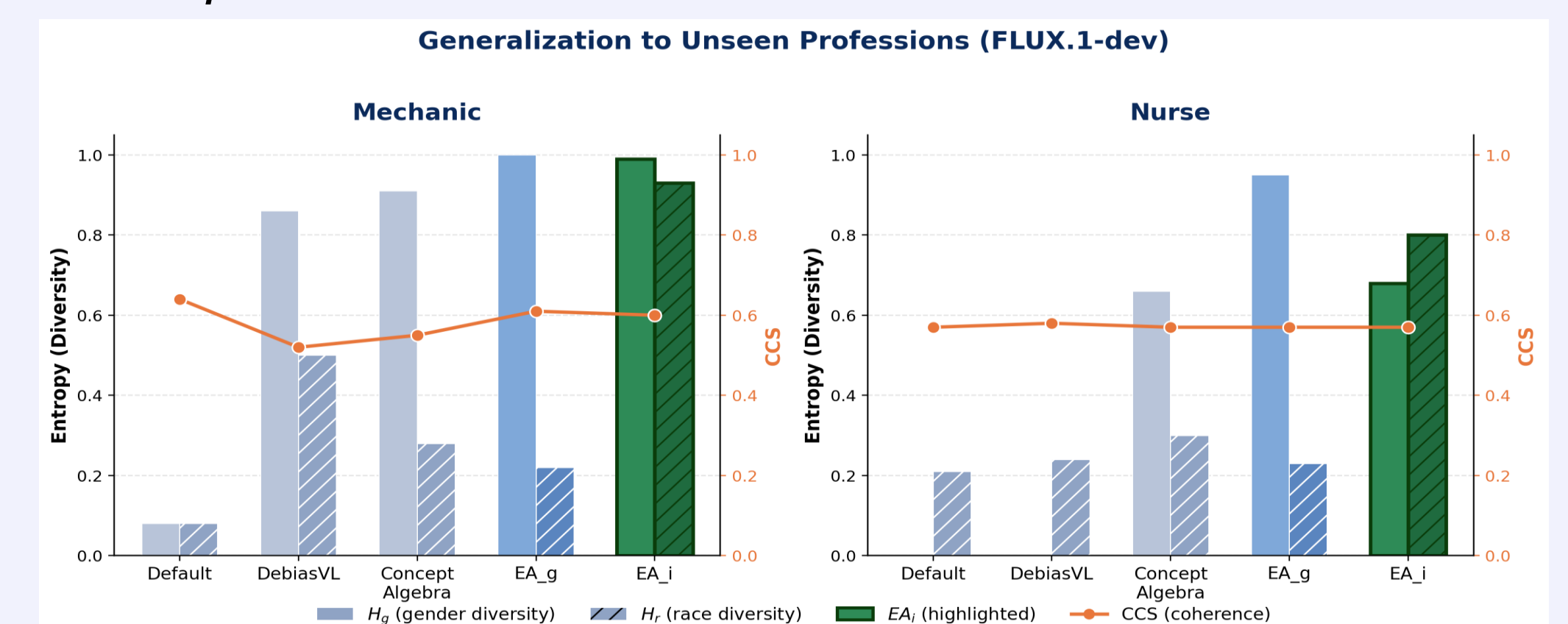
Hypothesis: The semantic manifold is locally linear, meaning that vector translations are effective and predictable within a bounded region around the base concept.



Gender confidence collapses early ($\alpha \approx 0.5 - 1.5$), while concept coherence (CCS) remains stable until $\alpha \approx 2.0$, revealing a safe operating window where the attribute flips without sacrificing image fidelity, before both degrade together at higher α .

Generalizability

Hypothesis: The generalized attribute vectors (v_A) are not overfitted to the source concepts used for deriving them and should generalize to a disjoint set of unseen concepts.



Our method (EA_i) generalizes to professions unseen when computing attribute vectors, delivering the largest gains in race diversity over other baselines (e.g., Mechanic: 0.22 $\rightarrow$ 0.93) without sacrificing concept coherence.

References

- Smith, M., Sambandham, V.T., Richter, F., Uhl, M., Crompton, L. and Schön, T., Operationalizing Fairness in Text-to-Image Models: A Survey of Bias, Fairness Audits and Mitigation Strategies. In *Algorithmic Fairness Across Alignment Procedures and Agentic Systems*.
- Bolukbasi, T., Chang, K.W., Zou, J.Y., Saligrama, V. and Kalai, A.T., 2016. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. *Advances in neural information processing systems*, 29.

